

Course Code: CS2306		
Name of the Programme: BTech (H)		
Name of the Course: Mobile Application Development Essentials		
Semester: IV		
Course credit	No. of hours per week (L + T + P)	Total no. of Teaching hours
0:1:2	0+15+30	45
Course Objectives:		
a) To introduce students to the mobile application ecosystem and app development lifecycle, including no-code, low-code, and traditional development approaches.		
b) To enable students to design and build functional mobile applications using no-code and low-code platforms.		
c) To provide guided exposure to Android Studio and real mobile app project structure through template-based and assisted development.		
d) To develop UI/UX design skills, product thinking, and basic knowledge of testing and deployment of mobile applications.		

Course outline (Syllabus of the course)

Syllabus:

Module 1: Mobile App Ecosystem & No-Code Paradigm	No. of hours
	7 Hours (3T + 4P)
Mobile apps vs Web apps, Native vs Cross-platform vs No-code, App development lifecycle, Overview of no-code tools (Glide / Thunkable / AppSheet / FlutterFlow), Use cases.	

Module 2: UI/UX Design & Visual App Building	No. of hours
	10 Hours (4T + 6P)
Screen flow, navigation, UI components, design principles, wireframing, mapping UI to app screens.	

Module 3: Data, Logic & Workflow (Low-Code Concepts)	No. of hours
	12 Hours (4T + 8P)
Tables, records, CRUD, workflows, visual logic, built-in databases, intro to APIs.	

Module 4: Android Studio Exposure (Assisted Development)	No. of hours
	10 Hours (2T + 8P)
Android Studio overview, project structure, activities, layouts, template-based app modification.	

Module 5: Testing, Packaging & Mini Project	No. of hours
	6 Hours (2T + 4P)
Testing, debugging, APK generation, Play Store overview, mini project development.	

LAB Experiments List

1. Explore No-Code Platform & Build Profile App
2. Build Multi-Screen Navigation App
3. Build Form-Based Data Collection App
4. Build CRUD App using No-Code / Low-Code Tool
5. Android Studio: Run & Modify Template App
6. Android Studio: Build Simple 2–3 Screen App
7. Mini Project: Real-world App Prototype

Each experiment should include: Aim, Tool used, Procedure, Output, Viva questions

Course outcomes	
a)	Understand the mobile app ecosystem and development approaches.
b)	Build functional apps using no-code and low-code tools.
c)	Apply UI/UX, data handling, and workflow logic in app development.
d)	Demonstrate basic understanding of Android app structure and deployment.

Text Books	
1	Build Apps Without Coding: A Beginner's Guide to No-Code Platforms, Alex Thompson, TechPress, 2024, ISBN: 978-1-23456-789-7

Reference books	
1	No-Code App Development: A Comprehensive Beginner's Guide to Building Robust Web and Mobile Applications Without Coding, Benjamin Rich, Independently Published, 2025, ISBN: 9798268937985
2	Prototyping and UX Design for Mobile Apps, Lisa Duncan & Marco Perez, DesignBooks Publishing, 2023, ISBN: 978-0-98765-432-1
3	Android Application Development All-In-One For Dummies, 3rd ed., Barry Burd, John Paul Mueller, Wiley, 2021, ISBN: 9354245781